

Individual Assessment Coversheet

To be attached to the front of the assessment.

Campus: Tyger Valley

Faculty: Information Technology

Module Code: ITCFA1-33

Group: 4

Lecturer's Name: Amakan Agoni

Student Full Name: Keenan Paiva Batista

Student Number: 4845350

Indicate	Yes	No
Plagiarism report attached		

Declaration:

I declare that this assessment is my own original work except for source material explicitly acknowledged. I also declare that this assessment or any other of my original work related to it has not been previously, or is not being simultaneously, submitted for this or any other course. I am aware of the AI policy and acknowledge that I have not used any AI technology to generate or manipulate data, other than as permitted by the assessment instructions. I also declare that I am aware of the Institution's policy and regulations on honesty in academic work as set out in the Conditions of Enrolment, and of the disciplinary guidelines applicable to breaches of such policy and regulations.

Signature 	Date 29 August 2024
---	-------------------------------

Lecturer's Comments:

--

Marks Awarded:	%
-----------------------	----------

Signature	Date
------------------	-------------

Table of Contents

Section A

Question 1.....Page 1-2

Question 2.....Page 3-4

Question 3.....Page 4-5

Question 4.....Page 5-6

Bibliography.....Page 7

Section A

Question 1

1.1) When migrating to the cloud, the business should make use of the AWS Cloud Adoption Framework (CAF) that indicates key elements that the organisation should take into consideration. The perspectives of this framework includes the business, people, governance, platform, security and operations perspective.

In terms of a business perspective, the organisation needs to ensure that migrating to the cloud will provide benefits and features are aligned with the requirements of the business. Therefore, in this scenario the business needs to ensure that the cloud will have global reach enabling employees to easily access company resources.

Moreover, a people perspective refers to all organisation structures and human resource capabilities that are required to ensure the successful adoption of cloud systems with personnel. This is accomplished by the organisation implementing training procedures for staff to effectively utilise cloud systems and ensuring that staff roles are properly assigned and managed.

In addition, the governance perspective requires the business to take into consideration how to monitor and manage the adoption of cloud systems across all sectors of the business to ensure that it is integrated properly. This involves the business creating a strategy of how to efficiently manage business processes such as projects and programs.

The platform perspective refers to creating procedures required to develop enterprise-grade cloud systems and introduce new solutions to the business's cloud systems. These procedures can consist of solution architecture and patterns in cloud adoption.

Furthermore, the security perspective describes the security measures that are required for protecting company resources over the cloud. This perspective is one of the important objectives that the business is interested in when migrating to cloud system and it can be solved with various cloud implementations. A private cloud deployment model can be considered for the business as it ensures that resources and services stored on the cloud can only be accessed by individuals in the organisation, providing enhanced security and allowing the organisation to easily stay compliant with industry standards.

Additionally, the operations perspective is also a component that the business is interested in achieving, because it refers to disaster recovery and the process of recovering workloads for the business to continue with operations. This perspective can be accomplished if the business decides to utilise an infrastructure as a service (IaaS) cloud consulting service delivery model, because it allows for the organisation to manage the cloud infrastructure and comes with the benefits of scalability as well as disaster recovery for company resources.

1.2) The migration to the cloud introduces various benefits such as additional cost savings for the business. Most cloud services operate with a pay-as-you-go pricing model, which allows the business to avoid paying upfront costs. Moreover, the business does not need to manage and maintain the physical infrastructure that hosts the cloud service as it is the responsibility of the cloud service provider.

Additionally, cloud services are also equipped with backup, recovery as well as failover features and systems. These features are often built into the cloud service for business to utilise and some cloud providers enable companies to store data backups across various geographical locations (Microsoft Azure, 2024).

Furthermore, the cloud can also improve the security to protect the company's resources. Many cloud services regularly receive software and security patches as well as come with security policies and tools that can further protect company resources.

In addition, cloud migration enables the organisation to more easily deploy and manage their resources globally. This benefit is possible due to the physical cloud infrastructure that cloud service providers have spread across globally, enabling employees and customers to have a smoother experience when accessing company resources and applications.

Moreover, cloud services can also simplify the management and monitoring of both on-premises and cloud resources. Cloud providers often provide a central management tool that enables the company that can be used to easily manage the company's physical infrastructure and cloud resources in a single place (Microsoft Azure, 2024).

1.3) Amazon Elastic File System (Amazon EFS) is an AWS storage service that offers an elastic Network File System that can be fully managed and is scalable. This service meets the requirement of slowly integrating the organisation's services to the cloud, because it can be used for resources on both the AWS cloud and company's on-premises infrastructure. This feature allows the organisation to securely store its data without fully migrating towards the cloud.

In addition, AWS Transit Gateway is an AWS network service that enables connections between networks through a single gateway. This proves to be beneficial to the organisation, because customers can connect their Amazon Virtual Private Cloud directly to the organisation's on-premises network. This allows the organisation to establish communication with its customers without having to fully integrate with the cloud.

Additionally, AWS Direct Connect is another AWS network service, which permits the organisation to create a private network connection between AWS and their premises. This smoothenes the transition towards the cloud system for the organisation, because it increases the bandwidth and makes the process less costly.

Moreover, AWS IAM is a security service offered by AWS that can be used to apply restrictions to the type of actions and services that accounts can utilise. This proves valuable to the smooth migration to the cloud system as the organisation can prevent its employees from using services that the organisation has not fully implemented. The organisation can control the pace of the cloud system integration.

Question 2

2.1)

[AWS Pricing Calculator](#) > [My Estimate](#)

My Estimate

Edit

Estimate summary

Info

Upfront cost

21,765.00 USD

Monthly cost

4,629.71 USD

Total 12 months cost

77,321.52 USD

Includes upfront cost

My Estimate

Duplicate

Delete

Move to

Create group

Add support

Add service

Find resources

< 1 >

☐	Service Name	Status	Upfront cost	Monthly cost	Descrip...	Region	Config Summary
☐	Amazon EC2	-	21,765.00 USD	1,813.76 USD	-	Asia Pacific (Tokyo)	Tenancy (Shared Instan...
☐	Amazon Simple Storage Service (S3)	-	0.00 USD	3.39 USD	-	Asia Pacific (Tokyo)	S3 Standard storage (5...
☐	Elastic Load Balancing	-	0.00 USD	54.17 USD	-	Asia Pacific (Tokyo)	Number of Application ...
☐	Amazon RDS for MySQL	-	0.00 USD	2,758.39 USD	-	Asia Pacific (Tokyo)	Storage amount (120 G...

2.2) The estimated annual costs is estimated to 77,321.52 USD and consists of the upfront cost as well as the monthly costs for a twelve-month period. The upfront cost is estimated to be 21,765.00 USD from the Amazon EC2 service. The monthly costs are estimated to 4629.71 USD for each month for all four services.

2.3) An Application Load Balancer (ALB) is a load balancer that assesses rules in the order of priority to decide the rule that needs to be applied, after it receives a request. It will then select a target from the target group for the rule that was applied. Features of ALB can include scalability for services by assigning it to an auto scaling group, improved overall load balancer performance as well as provides the ability for routing of requests to be based on the fields in the requests (AWS, 2024).

A Network Load Balancer (NLB) is a load balancer that uses the default rule and selects a target from target group, after it receives a connection request. It will then attempt to enable a TCP connection between the connection request and the port associated with the target selected.

Benefits of the NLB consist of its ability to scale in order to increase performance and handle larger amounts of requests per second as well as supporting static IP addresses, TCP and UDP traffic (AWS, 2024).

If the application is expected to receive many requests for connection that an NLB should be used due to its ability to handle millions of requests and the different connection types it can support. An ALB can be used to scenarios where requests are needed to be sent to multiple, specific applications due to how it can route requests to applications.

Question 3

3.1) Amazon Virtual Private Cloud (VPC) is the most appropriate AWS service for the organisation as it enables for AWS resources to be launched within a virtual network that is isolated from other virtual networks within AWS. Therefore, it enables the organisation to fully customise and configure the aspects associated with their virtual network.

Therefore, this level of customisation provided to the organisation gives them the ability to configure both the network gateways and route tables. Furthermore, the organisation also has full control over the IP addresses and its IP address range allowing them to effortlessly create subnets. These aspects are beneficial for the organisation, because it is the main requirements that they desired for their virtual networks.

However, AWS VPC has additional benefits and features that can prove to be valuable for the organisation. An instance of this is the ability to increase the security of the virtual network as network access control lists and security groups can be established. Further protection of the resources over the virtual network can be developed as both IPV4 and IPV6 can be used simultaneously. Moreover, the virtual network is scalable, since the AWS infrastructure can connect to the physical resources and the virtual network is similar to a traditional network. Scalability enables the organisation to increase the overall scale of the virtual network allowing for more resources to be utilised or decrease it to save on costs.

3.2) The first step to access the virtual private cloud (VPC) is to open the web interface provided by the AWS Management Console (AWS, 2024). Users can then search for the VPC dashboard in the search function, open the “Your VPCs” submenu and access the details of the VPC that they select within the submenu. The VPC details that are displayed when it is selected includes DNS hostnames and resolution, the VPC and owner ID as well as components of IPV4 and IPV6. Furthermore, it also provides the direction to important aspects of the VPC such as the DHCP option set, main route table and network ACL.

3.3) In terms of security, AWS makes use of a shared responsibility model where their customers are responsible for the protection of their data within the cloud environment and AWS ensures the protection of the cloud environment itself. AWS ensures that their physical infrastructure, that hosts their cloud services and allows other organisations to access it, is protected and properly maintained. The physical infrastructure includes data centres, AWS regions, availability zones as well as edge locations. Furthermore, AWS is also responsible for the software that controls their cloud services by providing security updates and patches.

In addition, AWS customers have access to security services provided by AWS enabling them to perform and maintain their own security measures and procedures. AWS Identity and Access Management (IAM) is an example of these security services where organisations are able to create and impose authorisation policies that determines which users will have access and permissions for specific AWS cloud services. This security service has additional functionalities such as the ability for the organisation to easily define the permissions for a group of users simultaneously, create a policy document that outlines user permissions as well as provide temporary permissions for a user.

Moreover, AWS implements security auditors that continuously checks if AWS is compliant with laws and regulations as well as inform both AWS and its customers of what they find, which will determine what decision they will need to take. AWS also offers cloud services that helps customers to stay compliant such as AWS Artifact, which is a service that delivers on-demand compliance reports from AWS and select agreements with AWS for present or future user accounts (AWS, 2024). AWS Lambda is another service that can be used by customers for compliance, because it can automatically fix any compliance problems that it detects without input from the organisation.

Question 4

4.1) Amazon Elastic Cloud Compute (EC2) is an AWS service that can create and run virtual machines, known as EC2 instances, where the majority of server operating systems are supported such as Windows 2008, 2012, Amazon Linux and Ubuntu. This service also offers flexibility through the ability for users to increase or decrease the number or size of servers that are used to host an application. Therefore, Amazon EC2 is the most appropriate AWS service that can be utilised to produce virtual servers that hosts the same applications on physical servers.

There are various benefits that are presented by the Amazon EC2 service like cost efficiency. Amazon provides and maintains the physical infrastructure needed to host this service and its features, while users can avoid upfront payments and follow a pay-as-you-go pricing model to save on costs. Furthermore, the EC2 instances that can be created from this service is extremely configurable and customisable. Each EC2 instance can be created from various instance types and operating systems as well as have a diverse amount of storage and memory that can fit the requirements for the user. Moreover, Amazon EC2 also implements scalability for each EC2 instance. This enables users to scale the capacity of instances at the same time depending on the situation.

4.2) The first step to hosting an application on through Amazon EC2 is access the AWS management console and from their search for the EC2 service in the search function. While on the EC2 dashboard, user must then click the button where they can launch an EC2 instance. The user must then configure the details of the EC2 instance, such as its name, operating system, instance type, network and storage settings, to what is most appropriate for the application that they want to host. After configuring all these details, the user can then create and launch the EC2 instance.

Afterwards, the user should then search for AWS CodeDeploy where they should create an application with a deployment group and specifying the EC2 instance that will be used to deploy the application. The user must then upload a revision of their application to Amazon S3 and then

deploy it in the same deployment group made from CodeDeploy. At this point in time the user will then be able to deploy and host their application on the EC2 instance with the use of the application revision in the deployment group (AWS, 2024).

4.3) In this scenario, a method of securing API calls to other AWS services is with the implementation of AWS Lambda. AWS Lambda is a compute service that can automatically respond to events and manage various computing tasks such as calling APIs and integrating with other AWS services. Therefore, AWS Lambda can be utilised in this scenario to automatically respond to any API requests from the EC2 instances and secure these API calls to other services, since AWS Lambda is integrated with the other services.

As an alternative to AWS Lambda, Amazon API Gateway can be used as it is an AWS service that can create, monitor and secure REST, HTTP and WebSocket APIs. Additionally, Amazon API Gateway enables APIs to access the different AWS services, and it utilises powerful authentication mechanisms offered by AWS services like AWS IAM to ensure the security of API calls. Furthermore, the efficiency of API calls are also improved by the API Gateway through proficient traffic management and access control (AWS, 2024). Therefore, the API Gateway can use its capabilities to secure the different types of APIs between several AWS services.

Bibliography

AWS, 2024. *AWS Artifact*. [Online]

Available at: <https://aws.amazon.com/artifact/?c=sc&sec=srvvm>

[Accessed 21 August 2024].

AWS, 2024. *Deployments on an EC2/On-Premises Compute Platform*. [Online]

Available at: <https://docs.aws.amazon.com/codedeploy/latest/userguide/deployment-steps-server.html>

[Accessed 29 August 2024].

AWS, 2024. *What is a Network Load Balancer?*. [Online]

Available at: <https://docs.aws.amazon.com/elasticloadbalancing/latest/network/introduction.html>

[Accessed 27 August 2024].

AWS, 2024. *What is Amazon API Gateway?*. [Online]

Available at: <https://docs.aws.amazon.com/apigateway/latest/developerguide/welcome.html>

[Accessed 29 August 2024].

AWS, 2024. *What is Amazon VPC?*. [Online]

Available at: <https://docs.aws.amazon.com/vpc/latest/userguide/what-is-amazon-vpc.html>

[Accessed 26 August 2024].

AWS, 2024. *What is an Application Load Balancer?*. [Online]

Available at: <https://docs.aws.amazon.com/elasticloadbalancing/latest/application/introduction.html>

[Accessed 27 August 2024].

Microsoft Azure, 2024. *Benefits of cloud migration*. [Online]

Available at: <https://azure.microsoft.com/en-us/resources/cloud-computing-dictionary/benefits-of-cloud-migration>

[Accessed 27 August 2024].